

From Proof-Nets to Interaction Nets

Yves Lafont

CNRS
Laboratoire de Mathématiques Discrètes *

June 13, 1994

1 Introduction

If we consider the interpretation of proofs as programs, say in intuitionistic logic, the question of equality between proofs becomes crucial: The syntax introduces meaningless distinctions whereas the (denotational) semantics makes excessive identifications. This question does not have a simple answer in general, but it leads to the notion of proof-net, which is one of the main novelties of linear logic. This has been already explained in [Gir87] and [GLT89].

The notion of interaction net introduced in [Laf90] comes from an attempt to implement the reduction of these proof-nets. It happens to be a simple model of parallel computation, and so it can be presented independently of linear logic, as in [Laf94]. However, we think that it is also useful to relate the exact origin of interaction nets, especially for readers with some knowledge in linear logic. We take this opportunity to give a survey of the theory of proof-nets, including a new proof of the sequentialization theorem.

2 Multiplicatives

First we consider the kernel of linear logic, with only two connectives: $\otimes$ (*times* or *tensor product*) and its dual $\wp$ (*par* or *tensor sum*). The first one can be seen as a conjunction and the second one as a disjunction. Each atom has a positive form p and a negative one $p^\perp$ (the *linear negation* of p). This linear negation is extended to all formulae according to the following laws:

$$p^{\perp\perp} = p, \quad (A \otimes B)^\perp = B^\perp \wp A^\perp, \quad (A \wp B)^\perp = B^\perp \otimes A^\perp.$$

Note that our negation reverses the order of subformulae, although it would only matter in the case of non-commutative linear logic. In the present case, a sequent $\vdash A_1, \dots, A_n$ is a multiset, which

*Address: 163 avenue de Luminy, case 930, F 13288 MARSEILLE CEDEX 9. Email: lafont@lmd.univ-mrs.fr

means that the order of formulae is irrelevant. With this convention, there are only four deduction rules (*axiom*, *cut*, *times* and *par*):

$$\frac{}{\vdash A, A^\perp} \quad \frac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} \quad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \quad \frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma}$$

This sequent calculus satisfies the cut-elimination property, which means that any proof of a sequent $\vdash \Gamma$ reduces to a cut-free one. For instance, the sequent $\vdash p^\perp, p \otimes (q \otimes r), r^\perp \wp q^\perp$ has two *cut-free* proofs:

$$\frac{\frac{\frac{}{\vdash p^\perp, p} \quad \frac{\frac{\frac{}{\vdash q, q^\perp}}{\vdash q \otimes r, r^\perp, q^\perp} \quad \frac{}{\vdash r, r^\perp}}{\vdash q \otimes r, r^\perp, q^\perp}}{\vdash p^\perp, p \otimes (q \otimes r), r^\perp, q^\perp}}{\vdash p^\perp, p \otimes (q \otimes r), r^\perp \wp q^\perp} \quad \text{and} \quad \frac{\frac{\frac{}{\vdash p^\perp, p} \quad \frac{\frac{\frac{}{\vdash q, q^\perp}}{\vdash q \otimes r, r^\perp, q^\perp} \quad \frac{}{\vdash r, r^\perp}}{\vdash q \otimes r, r^\perp, q^\perp}}{\vdash p^\perp, p \otimes (q \otimes r), r^\perp \wp q^\perp}}{\vdash p^\perp, p \otimes (q \otimes r), r^\perp \wp q^\perp}$$

By cut-elimination, any proof of this sequent reduces to one of those two proofs. There are three basic reduction rules for cut-elimination:

$$\frac{\frac{}{\vdash A^\perp, A} \quad \frac{\vdots}{\vdash A, \Gamma}}{\vdash A, \Gamma} \longrightarrow \frac{\vdots}{\vdash A, \Gamma} \quad \frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{}{\vdash A^\perp, A}}{\vdash A, \Gamma} \longrightarrow \frac{\vdots}{\vdash A, \Gamma}$$

$$\frac{\frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{\vdots}{\vdash B, \Delta}}{\vdash A \otimes B, \Gamma, \Delta} \quad \frac{\frac{\vdots}{\vdash B^\perp, A^\perp, \Lambda}}{\vdash B^\perp \wp A^\perp, \Lambda}}{\vdash \Gamma, \Delta, \Lambda} \longrightarrow \frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{\frac{\vdots}{\vdash B, \Delta} \quad \frac{\vdots}{\vdash B^\perp, A^\perp, \Lambda}}{\vdash A^\perp, \Delta, \Lambda}}{\vdash \Gamma, \Delta, \Lambda}$$

The other reduction rules allow to move cuts upwards, as in the following case:

$$\frac{\frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{\vdots}{\vdash B, C, \Delta}}{\vdash A \otimes B, C, \Gamma, \Delta} \quad \frac{\vdots}{\vdash C^\perp, \Lambda}}{\vdash A \otimes B, \Gamma, \Delta, \Lambda} \longrightarrow \frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{\frac{\vdots}{\vdash B, C, \Delta} \quad \frac{\vdots}{\vdash C^\perp, \Lambda}}{\vdash B, \Delta, \Lambda}}{\vdash A \otimes B, \Gamma, \Delta, \Lambda}$$

This reduction terminates but is not confluent. In other words, the normal form of a proof is not unique in general. This comes from the fact that, in sequent calculus, the order of application of rules may be irrelevant. *Proof-nets* allow one to abstract from this irrelevant order.

Beside cut-elimination, there is another transformation for eliminating non-atomic axioms. It

is an expansion rather than a reduction:

$$\frac{}{\vdash A \otimes B, B^\perp \wp A^\perp} \longrightarrow \frac{\frac{}{\vdash A, A^\perp} \quad \frac{}{\vdash B, B^\perp}}{\vdash A \otimes B, B^\perp, A^\perp} \quad \frac{}{\vdash A \otimes B, B^\perp \wp A^\perp}$$

This means that, if a sequent is provable, it has a cut-free proof with only atomic axioms.

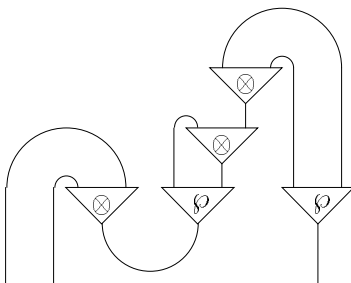
3 Proof-nets

Since our notion differs slightly from the original one in [Gir87], we shall give precise definitions. First, we introduce a *times cell* and a *par cell*:



Both have three *ports*: a *principal* one below and two *auxiliary* ones above. The two auxiliary ports are distinguished: There is a left one and a right one.

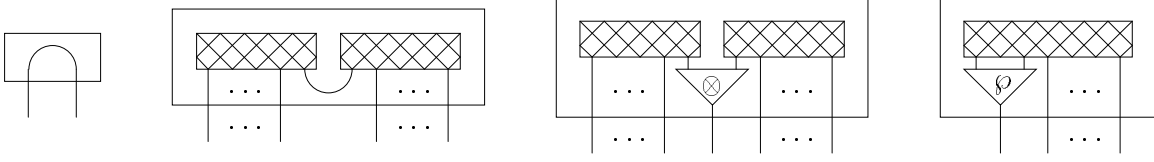
A *net* is just a finite graph built with times and par cells. In addition to the ports associated with those cells, there is an extra set of *free ports* for subsequent connections, and each port must be connected to another one by a *wire*. For instance, here is a net with 3 free ports:



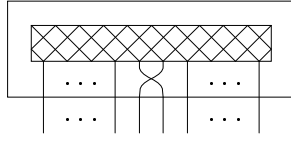
We say that a net is *well-formed* if it is of one of the following forms:

- a single wire,
- two well-formed nets connected by a single wire,
- two well-formed nets connected by a times cell through its auxiliary ports,
- a well-formed net connected to itself by a par cell through its auxiliary ports.

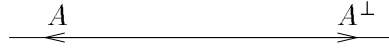
This is of course an inductive definition. Intuitively, a well-formed net is a kind of module obtained by plugging smaller modules according to certain rules. The free ports are the interface of this module. The four cases can be pictured as follows:



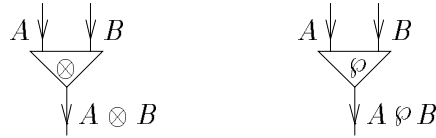
For instance, the above net is well-formed (see figure 1). In fact, the four cases correspond to the four rules of sequent calculus. Since we do not impose the planarity of nets, wires are allowed to cross each other, and the exchange rule is implicit:



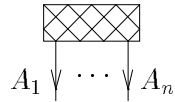
A *typed net* is a net where each wire is labelled with a formula. If a wire is explicitly labelled with A for a given orientation, then it is implicitly labelled with $A^\perp$ for the opposite orientation:



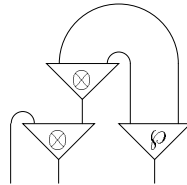
Furthermore, for each cell, the connecting wires must be typed as follows:



For instance, our favorite net can be typed (see figure 2): It defines a *proof-net*, *i.e.* a well-formed typed net. Clearly, a proof Π of a sequent $\vdash A_1, \dots, A_n$ defines a proof-net $\langle \Pi \rangle$ with n free ports typed as follows:



Conversely, any proof-net π is obtained in this way, although not uniquely. For instance, the two cut-free proofs of the sequent $\vdash p^\perp, p \otimes (q \otimes r), r^\perp \wp q^\perp$ define the same net:



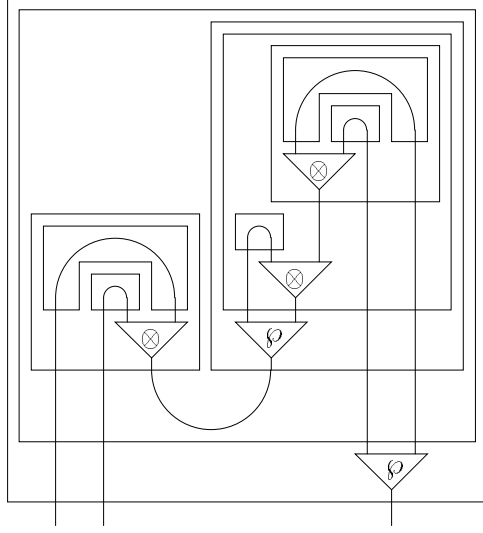


Figure 1: checking that a net is well-formed

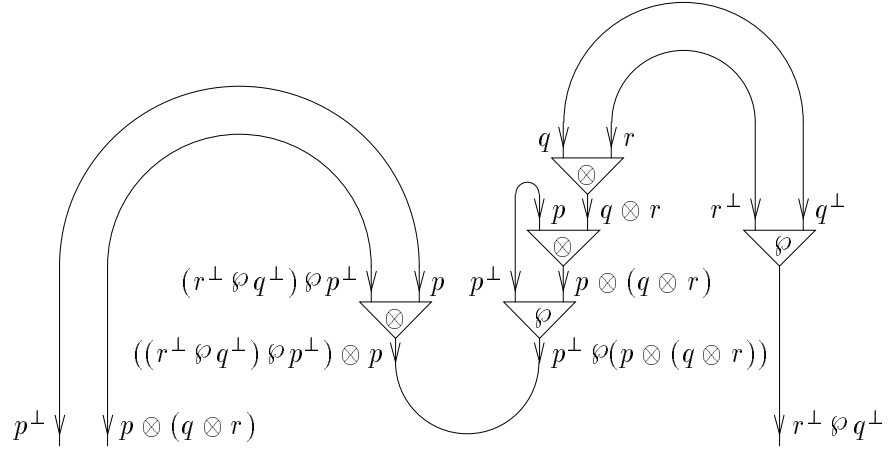


Figure 2: typing a net

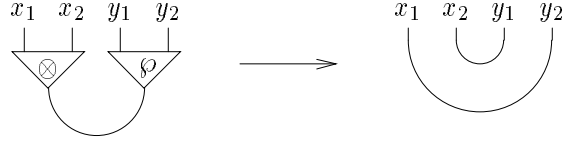
This means that the proof-net is more intrinsic than the proof. In fact, it is possible to show that two cut-free proofs define the same net if and only if they are equivalent modulo some permutations of rules of the following kind:

$$\frac{\frac{\vdots}{\vdash A, \Gamma} \quad \frac{\vdots}{\vdash B, C, D, \Delta}}{\vdash A \otimes B, C, D, \Gamma, \Delta} \sim \frac{\vdots}{\vdash A, \Gamma} \quad \frac{\vdots}{\vdash B, C, D, \Delta} \quad \frac{\vdots}{\vdash B, C \wp D, \Gamma, \Delta} \quad \frac{\vdots}{\vdash A \otimes B, C \wp D, \Gamma, \Delta}$$

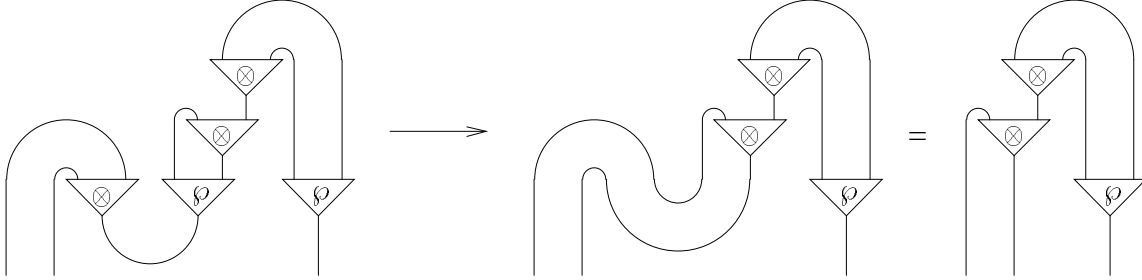
However, it is much better to think in terms of proof-nets than in terms of equivalence classes of proofs.

4 Reducing

With nets, the cut-elimination becomes surprisingly simple. There is only one reduction rule:



This means that, if the principal port of a times cell is connected to the principal port of a par cell, the two cells disappear and the remaining wires are joined together as indicated by the right member of the rule. Here is an example of reduction:



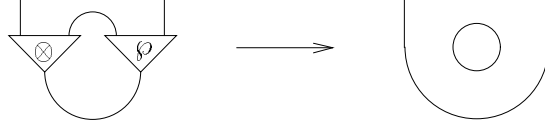
Of course, the reduction terminates since it decreases the number of cells, but it is also confluent in a strong sense:

Lemma 1 *If a net π reduces in one step to π' and to π'' , with $\pi' \neq \pi''$, then π' and π'' reduce in one step to a common net.*

The point is that, in an instance of the left member of the rule, the two cells are connected through their principal ports, so that two such instances are necessarily disjoint. As a corollary, the number

of steps for reducing a net to its normal form is independent of the reduction path. Note that our notion of net avoids the critical pairs of [Gir87] because we have no explicit cell for axiom and cut.

To be honest, there is a little problem with our definition. Indeed, if we apply the reduction rule to an arbitrary net, we may have to connect a wire with itself as in the following example:



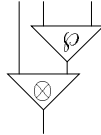
To fix this problem, we must allow *closed wires* in the definition of nets. Fortunately, we shall see that those closed wires cannot appear if we start from a well-formed net.

Clearly, if a proof Π reduces to Π' , then the net $\langle \Pi \rangle$ reduces to $\langle \Pi' \rangle$. In particular, if Π' is cut-free, then $\langle \Pi' \rangle$ is the normal form of $\langle \Pi \rangle$, which is uniquely determined by Π . Moreover, this normal form depends really on the proof, not only on the sequent. For instance, there are two distinct normal proof-nets for the sequent $\vdash p \otimes p, p^\perp, p^\perp$:



Conversely, it is possible to show that if $\langle \Pi \rangle$ reduces to a net π' , then there is a proof Π' such that Π reduces to Π' and $\pi' = \langle \Pi' \rangle$. In particular, proof-nets are closed under reduction.

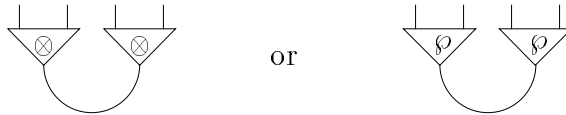
Now, we look more closely at the structure of irreducible nets. If $A[p_1, \dots, p_n]$ is a formula where the atoms $p_1, \dots, p_n$ occur positively, exactly once and in order, it defines a net $\langle A \rangle$ (the *tree* of A) with $n + 1$ free ports. For instance, if A is $p_1 \otimes (p_2 \wp p_3)$, the tree $\langle A \rangle$ is the following net:



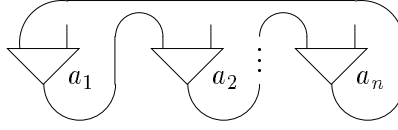
Note that, in general, $\langle A \rangle$ is not a proof-net, but it is clearly typable.

Lemma 2 *If π is an irreducible net, then at least one of the following statements holds:*

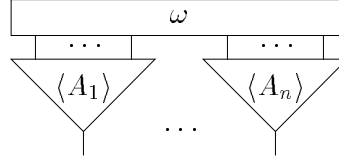
- π contains a *clash*, i.e. two cells connected through their principal ports for which no rule applies:



- π contains a vicious circle, i.e. a sequence of cells $a_1, a_2, \dots, a_n, a_{n+1} = a_1$ such that the principal port of a_i is connected to an auxiliary port of a_{i+1} :



- π is in reduced form, i.e. it can be uniquely decomposed as follows:



where $A_1, \dots, A_n$ is a sequence of formulae and ω is a net without cell and without closed wire, i.e. a partition of the remaining ports into pairs.

Here, a closed wire is considered as a vicious circle of length $n = 0$. This lemma is proved by induction on the number of cells. The point is that, if π contains some cell but no clash and no vicious circle, then some free port must be connected to the principal port of a cell.

It is clear that a typed net contains no clash, and it is easy to see that a well-formed net contains no vicious circle.¹ Therefore, an irreducible proof-net is necessarily in reduced form. Conversely, if π is in reduced form, then π is typable although not necessarily well-formed: The types are obtained by unifying the atoms in $A_1, \dots, A_n$ according to ω .

5 Criterion

Clearly, a net built only with times cells is well-formed if and only if it defines a connected acyclic graph, also called a tree. In general, it is still true that a well-formed net must be connected, but the acyclicity condition is neither necessary nor sufficient. Consider indeed the following nets:



The first one is well-formed but not acyclic and the second one is acyclic but not well-formed. This means that an alternative condition is needed. A *switching* of a net is a graph obtained by replacing

¹In fact, it also happens that a typed net contains no vicious circle of length $n > 0$, but it may contain closed wires.

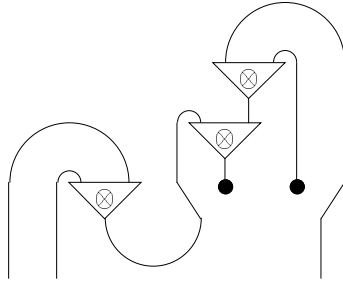
each par cell with one of the following two configurations:



For the two examples above, we get the following graphs:



In the first case, both are connected and acyclic, whereas in the second case, none is connected. All four switchings of our favorite proof-net are connected and acyclic. Here is one of them:



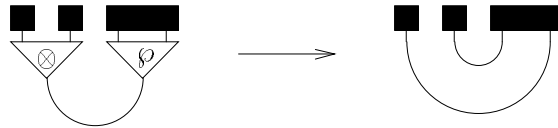
Theorem 1 *A net is well-formed if and only if every switching defines a connected acyclic graph.*

This is the *Danos-Regnier criterion* introduced in [DR89]. In the case of a typed net, we get the sequentialization theorem of [Gir87].

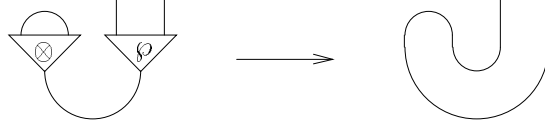
The proof of theorem 1 will be the object of the next section, but we can already make some comments. First, notice that if a net satisfies the criterion, then it contains no vicious circle: This is a direct consequence of the acyclicity condition. Moreover, if a net contains p par cells, it has 2^p switchings. Hence, the criterion itself does not provide an efficient algorithm for checking if a net is well-formed, but it is interesting from a theoretical viewpoint, because of its geometrical nature. For example, it is obviously preserved by reduction:

Lemma 3 *If π reduces to π' and π satisfies the criterion, so does π' .*

Indeed, let λ be a subnet of π consisting of a times cell and a par cell connected through their principal ports. The criterion ensures that any switching of the complement of λ consists of three connected components, two of them being connected to the times cell, and the third one to the par cell. If we apply the reduction to λ , we get a connected acyclic switching:



The lemma follows easily. By theorem 1, this means that well-formed nets are closed under reduction, and similarly for proof-nets, since typing is obviously preserved by reduction. Note that the converse of the lemma does not hold: π does not necessarily satisfy the criterion if π' does. Here is a simple counterexample:

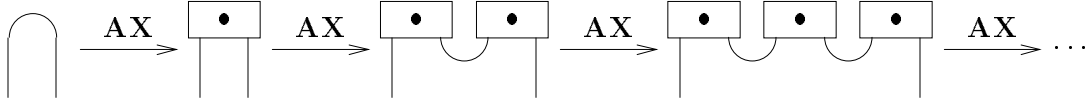


6 Parsing

Typability of nets can be checked with a very simple unification algorithm. To check if a net is well-formed, there is of course a naive exponential algorithm which tries all possibilities, but Danos and Regnier noticed that this can be done in quadratic time (see also [Gal91]). In order to show this, we introduce the *parsing box*:



This is a special kind of cell with a variable number of non-distinguished ports. Clearly, a net is well-formed if and only if it reduces to a parsing box by the rules of figure 3: This is just a reformulation of the definition. Of course, this reduction has nothing to do with cut-elimination! Unfortunately, it does not terminate:



To avoid this problem, we exclude the degenerate case of a net without cell:

Lemma 4 *A non-degenerate net is well-formed if and only if it reduces to a parsing box by the rules of figure 4.*

Proof: First notice that **times** is derivable from **TIMES** and **AX**. Similarly, **par** is derivable from **PAR** and **AX**. This shows that a net which reduces to a parsing box by the rules of figure 4 is well-formed. The converse is proved by case analysis, using in particular the fact that **TIMES** is derivable from **times** and **CUT**. **Q.e.d.**

Starting from a net with n cells, the parsing terminates in at most $2n - 1$ steps. Furthermore, it is confluent (see figure 5), and so it gives an efficient algorithm testing if a net is well-formed. In case of success, and if the net is typable, a proof of sequent calculus can be extracted from the reduction.

We can use this algorithm to prove theorem 1. First, we notice that the Danos-Regnier criterion makes sense in the case of nets with parsing boxes, and it is invariant by the parsing rules:

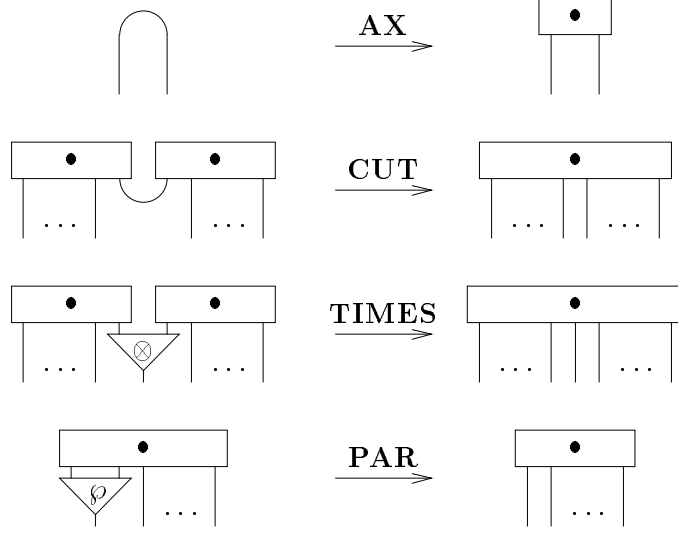


Figure 3: grammar for well-formed nets

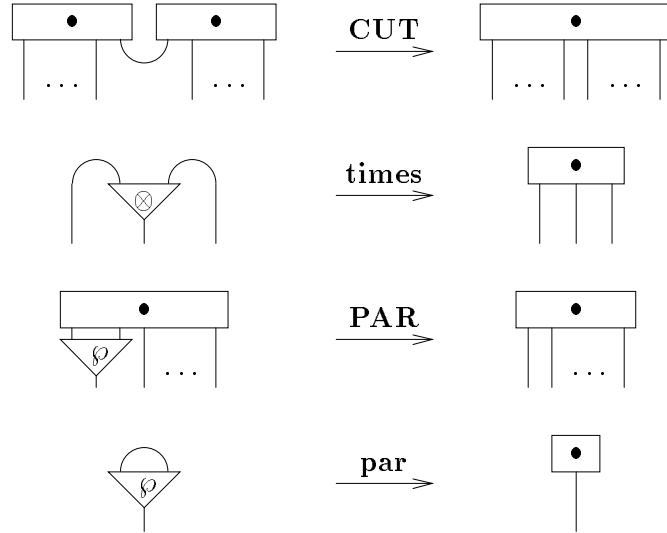


Figure 4: grammar for non-degenerate well-formed nets

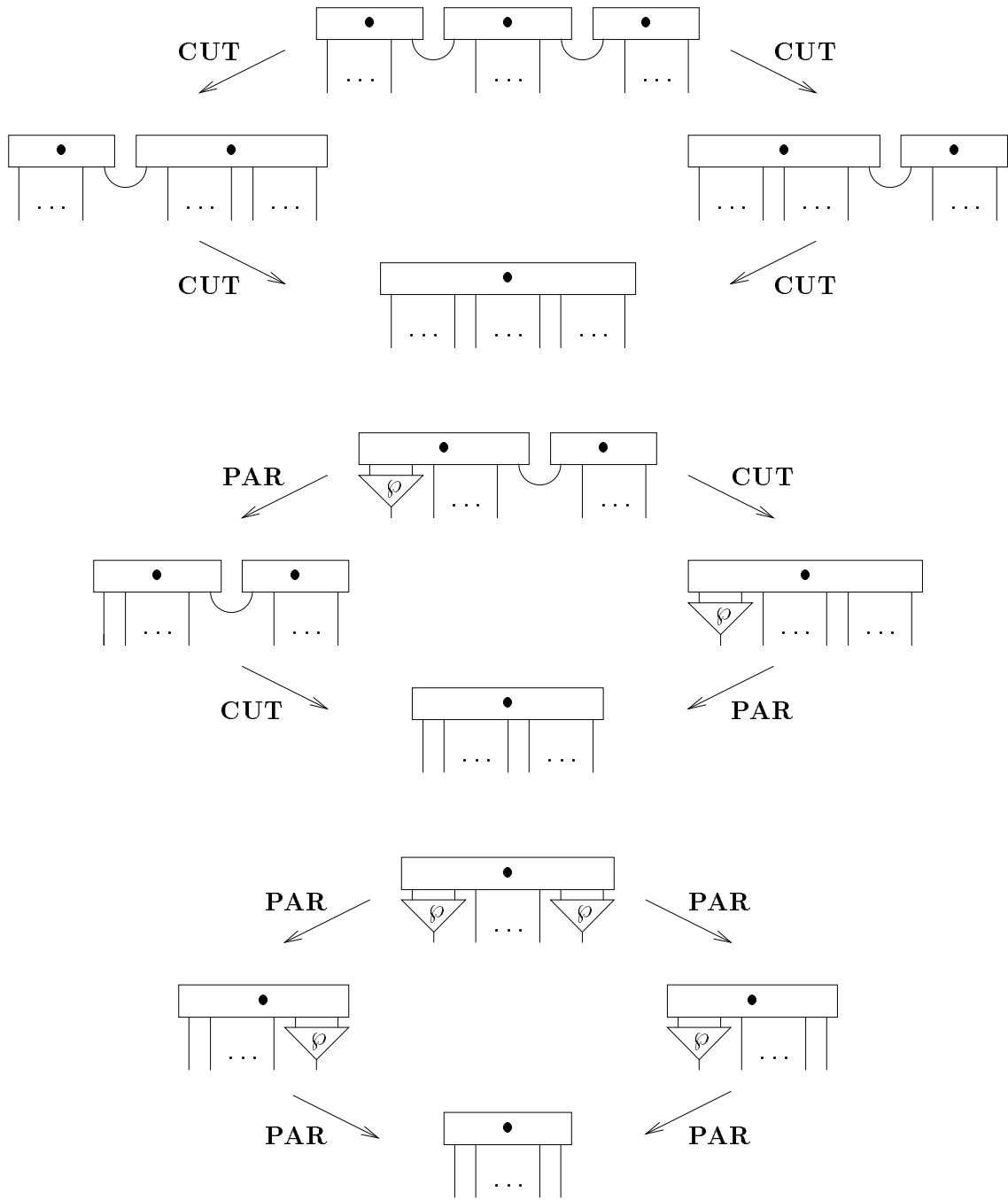


Figure 5: confluence of parsing

Lemma 5 *If π reduces to π' by one of the rules of figure 4, then π satisfies the criterion if and only if π' does.*

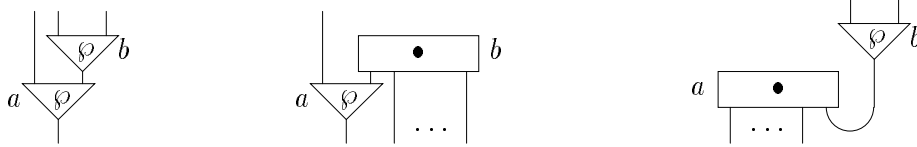
Proof: For any of the four rules, every switching of the left member is a tree, and similarly for the right member. Now, replacing a subtree by a tree in a tree yields a tree. **Q.e.d.**

The theorem follows from lemmas 4, 5 and the following argument, which was first pointed out in [Dan90]:

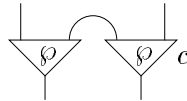
Lemma 6 *If a non-degenerate net π satisfies the Danos-Regnier criterion and is irreducible by the rules of figure 4, then π is a parsing box.*

Proof: Since π is irreducible by **times**, it contains no times cell. If a and b are cells in π , we say that a is *dominated by* b and we write $a \prec b$ when one of the following cases holds:

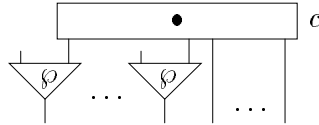
- a and b are par cells and an auxiliary port of a is connected to the principal port of b ,
- a is a par cell, b a parsing box and an auxiliary port of a is connected to a port of b ,
- a is a parsing box, b a par cell and a port of a is connected to the principal port of b .



By hypothesis, π contains at least one cell, and there is no cycle $a_1 \prec a_2 \prec \dots \prec a_{n+1} \prec a_n = a_1$, since it would give a cyclic switching. Hence there is a cell c which is dominated by no other one. Assume that c is a par cell and consider an auxiliary port of c . It cannot be connected to a free port, since it would give a non connected switching. Knowing that c is dominated by no other cell and that π is irreducible by **par**, we must have the following configuration:



But again, this configuration gives a non connected switching, hence c is a parsing box. Knowing that π is irreducible by the parsing rules and that c is dominated by no other cell, it can only be connected to free ports and to auxiliary ports of distinct par cells:



But in such a case, there is a non connected switching, unless c is only connected to free ports, which means that π is a parsing box. **Q.e.d.**

7 Units

Until now, we have not considered the multiplicative units **1** and **−** (*bottom*), with the following deduction rules:

$$\frac{}{\vdash \mathbf{1}} \qquad \frac{\vdash \Gamma}{\vdash -, \Gamma}$$

All statements of section 2 to 4 extend to the full multiplicative fragment. We just need a *one* cell and a *bottom* cell:



The reduction is very simple, with an empty right member:



Of course, it is no more the case that well-formed nets are connected. Moreover, the parsing rule for the bottom cell is strongly non-deterministic:



By this rule indeed, the bottom cell is allowed to interact with any parsing box in the net! In fact, there is no hope to find a good parsing algorithm for the full multiplicative fragment. In [LW94], it is shown indeed that the multiplicative fragment of linear logic with units but without proper atoms is NP-complete. But in the absence of proper atoms, a sequent $\vdash A_1, \dots, A_n$ is provable if and only the sequence of trees $\langle A_1 \rangle, \dots, \langle A_n \rangle$ defines a well-formed net. Therefore, parsing with units is NP-complete!

Moreover, there is no hope to extend the Danos-Regnier criterion to the full fragment. Indeed, a condition of the kind “for every switching ...” is co-NP, and by the previous remark, the existence of a co-NP condition would mean that $\text{NP} = \text{co-NP}$, which is very unlikely. For this reason, it is usually thought that something goes wrong with units. Now, there are two ways to escape: change the logic, or change the notion of net.

One can argue indeed that, in the criterion, acyclicity is more important than connectedness. For example, it is enough for preventing vicious circles. This suggests a weak version of the criterion, corresponding to a variant of linear logic with two extra deduction rules (*mix* and *empty*):

$$\frac{\vdash \Gamma \quad \vdash \Delta}{\vdash \Gamma, \Delta} \qquad \frac{}{\vdash}$$

This logic satisfies cut-elimination (see [FR90]), but it looks a bit degenerate. For instance, $A \otimes B$ implies $A \wp B$ and **1** is equivalent to **−**. On the other side, the proof-boxes of [Gir87] introduce a kind of synchronization which is not justified from a purely computational viewpoint. Maybe, a better understanding of linear logic is needed to settle this question.

8 Exponentials

We have seen that the process of cut-elimination in the sequent calculus for multiplicative linear logic can be completely *localized*. The contexts Γ , Δ and Λ play indeed no active role in the reduction rule. Unfortunately, the multiplicative fragment has very little computational power: The reduction terminates in linear time because of the absence of a contraction rule.

To gain expressiveness, we must also consider the *exponential* connectives: $!$ (*of course*) and its dual Γ (*why not*). There are four deduction rules (*promotion*, *dereliction*, *contraction* and *weakening*):

$$\frac{\vdash A, \Pi\Gamma}{\vdash !A, \Pi\Gamma} \quad \frac{\vdash A, \Gamma}{\vdash \Gamma A, \Gamma} \quad \frac{\vdash \Gamma A, \Gamma A, \Gamma}{\vdash \Gamma A, \Gamma} \quad \frac{\vdash \Gamma}{\vdash \Gamma A, \Gamma}$$

In the promotion rule, $\Pi\Gamma$ stands for a multiset $\Gamma C_1, \dots, \Gamma C_n$. Intuitively, the formula $!A$ represents an inexhaustible resource: It can be accessed (by dereliction), duplicated (by contraction) or erased (by weakening). The promotion rule expresses that a resource coming from inexhaustible resources is also inexhaustible.

This extended calculus satisfies the cut-elimination property, but the reduction of a proof may be quite long (hyperexponential in fact) because of the following reduction rule which duplicates proofs:

$$\frac{\frac{\vdots}{\vdash A, \Pi\Gamma} \quad \frac{\vdots}{\vdash \Gamma A^\perp, \Gamma A^\perp, \Delta}}{\vdash !A, \Pi\Gamma} \quad \frac{\vdots}{\vdash \Gamma A^\perp, \Delta} \quad \frac{\vdots}{\vdash A, \Pi\Gamma} \quad \frac{\frac{\vdash A, \Pi\Gamma}{\vdash !A, \Pi\Gamma} \quad \frac{\vdots}{\vdash \Gamma A^\perp, \Gamma A^\perp, \Delta}}{\vdash \Gamma A^\perp, \Pi\Gamma, \Delta} \quad \frac{\vdots}{\vdash \Pi\Gamma, \Pi\Gamma, \Delta} \quad \frac{\vdots}{\vdash \Pi\Gamma, \Delta}$$

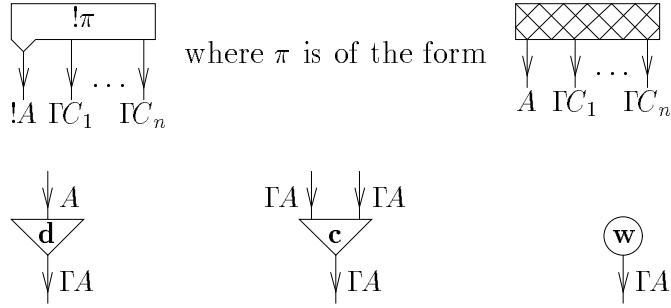
Here, the context $\Pi\Gamma$ of the promotion rule plays an active role in the reduction. For that reason, we introduce an *exponential box* for promotion and three ordinary cells for dereliction, contraction and weakening:



The exponential box is a special kind of cell parametrized by a net: If π is a net with $n + 1$ free ports, then $!\pi$ is a cell with one principal port (the leftmost one) and n auxiliary ports. The net π may itself contain exponential boxes.

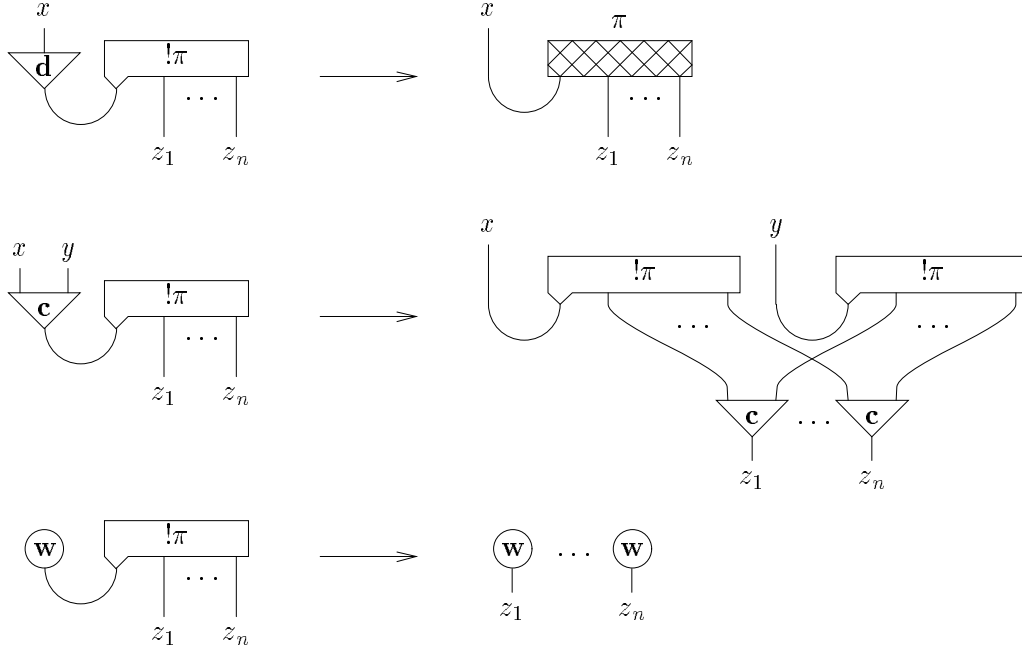
Following the rules of sequent calculus, it is easy to define an appropriate notion of well-formed net. For example, the exponential box $!\pi$ is well-formed when the net π is well-formed. The

contraction behaves like a *par* rule, and the weakening like a *bottom* rule. In particular, the Danos-Regnier criterion works only in the absence of weakening.² Finally, the typing constraints are the following:



Now we can express cut-elimination directly at the level of nets. Three kinds of reductions are needed:

- *external reductions* (when two cells or boxes are connected through their principal ports):

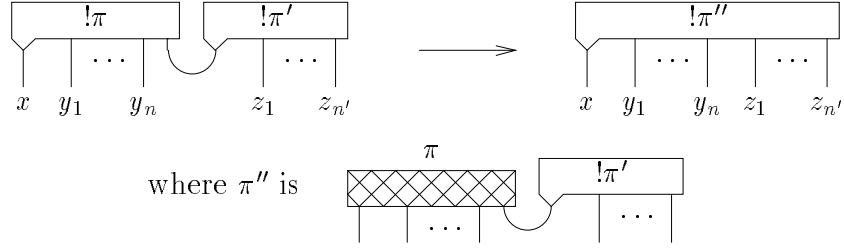


Intuitively, a dereliction opens a box, a contraction duplicates it and a weakening erases it.

- *commutative reductions* (when an auxiliary port of a box is connected to the principal port

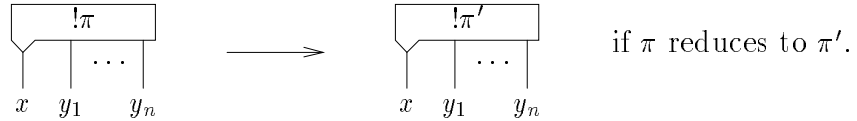
²For that reason, there is a box for weakening in [Gir87].

of another box):



Intuitively, the second box enters the first one.

- *internal reductions* (inside a box):



Of course, we must not forget the multiplicative reduction, which is of the external kind. This system terminates in the case of proof-nets (see [Gir87]). Furthermore, there are a lot of conflicts between the various kinds of rules, but the property of confluence holds, even in the untyped case (see [Dan90] or [Reg92]).³

External reduction suffices if we content ourselves with a weak notion of normal form. Here is another way of expressing this fact:

Lemma 7 *If Π is a proof of $\vdash \Gamma$, where Γ has no occurrence of the connective $!$, then the proof-net $\langle \Pi \rangle$ reduces to its normal form by external reduction.*

Indeed, if Γ has no occurrence of $!$ and $\langle \Pi \rangle$ is irreducible by external reduction, then it contains no box at all: This is easily proved by induction on Π . There is a similar situation in functional programming, where nobody wants to see actual values of functional type. In other words, external reduction is a kind of lazy reduction (see [Abr93]). Note that external reduction is confluent in the strong sense of lemma 1.

From the viewpoint of cut-elimination, this fragment is much more expressive than the multiplicative one. There is indeed a translation of the implicative fragment of intuitionistic logic into the multiplicative exponential fragment of linear logic (see [Gir87]). It transforms an intuitionistic proof of $\Gamma \vdash A$ into a linear proof of $\vdash \Pi\Gamma, A$ with the following interpretation of the implication:

$$A \Rightarrow B = \Gamma A^\perp \wp B.$$

This translation extends to untyped λ -terms (see [Dan90] or [Reg92]) so that β -reduction is interpreted in terms of reduction of well-formed nets⁴. There is another well-known translation

³This is not obvious at all, since for instance, the system for additive connectives is not confluent!

⁴Strictly speaking, this interpretation is only defined modulo associativity and unit axioms for contraction and weakening.

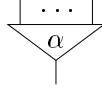
which transforms an intuitionistic proof of $\Gamma \vdash A$ into a linear proof of $\vdash \Gamma\Gamma, !A$ with the following interpretation of the implication:

$$A \Rightarrow B = \Gamma A^\perp \wp !B.$$

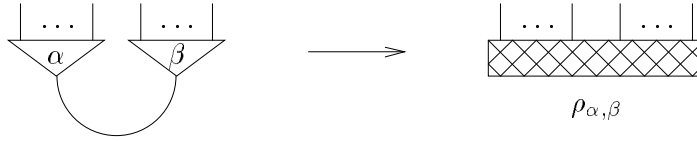
The difference is that, in the first case, external reduction corresponds to the call-by-name strategy, whereas in the second case, it corresponds to the call-by-value strategy. In both cases, an unsolvable term such as $(\lambda x \cdot xx)(\lambda x \cdot xx)$ gives a (well-formed but untyped) net with no normal form, even for external reduction.

9 Interaction nets

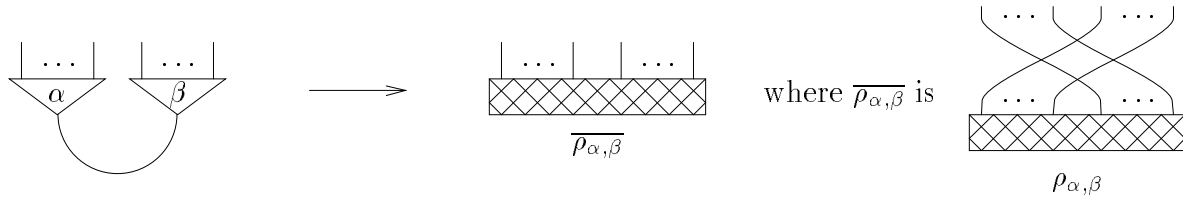
The previous sections suggest a natural generalization of proof-nets. Consider indeed an alphabet Σ of symbols with arities. For each symbol α of arity p in Σ , we can introduce a cell with one principal port and p auxiliary ones:



If α and β are symbols of respective arities p and q , an *interaction rule* for (α, β) is a reduction of the following form:



where $\rho_{\alpha, \beta}$ is a net with $p + q$ free ports, built with symbols of Σ . Obviously, this rule is equivalent to the following one:



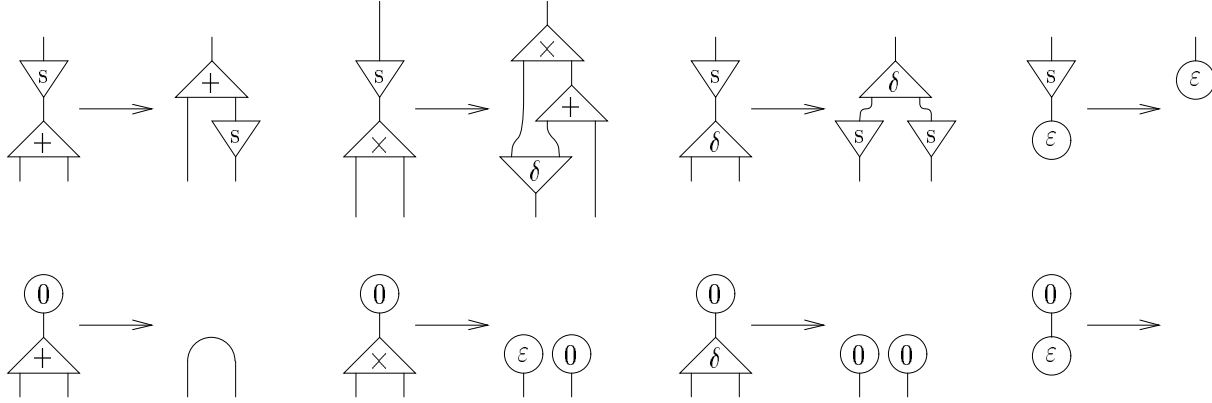
An *interaction system* is a non-ambiguous set of interaction rules. This means that the following conditions are satisfied:

- If a rule is given for (α, β) , then no other rule is given for (α, β) or (β, α) .
- If a rule is given for (α, α) then $\rho_{\alpha, \alpha} = \overline{\rho_{\alpha, \alpha}}$ (symmetry condition).⁵

⁵This case was not considered in [Laf90].

Such a system satisfies the strong confluence property of lemma 1.

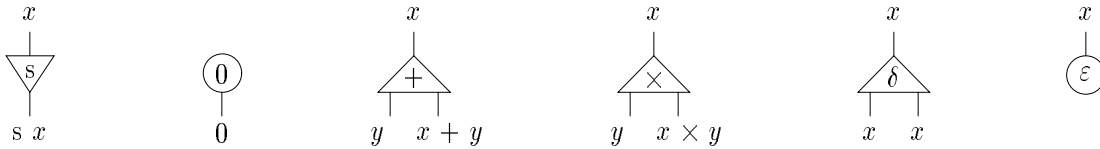
The multiplicative reduction of section 4 is a very simple kind of interaction rule where the net $\rho_{\alpha,\beta}$ contains no cell. A much more elaborate example is given by the external reduction of the previous section. In that case, the alphabet is infinite because there are infinitely many exponential boxes, but the system is *locally finite* in the following sense: If we start from a net π , the only cells that may appear in the reduction are those which occur in the construction of π (possibly inside boxes), and they are finitely many. In fact, it is possible to replace this infinite system by a finite one at the expense of some encoding (see [Mac94]).⁶ Here is a completely different kind of example:



It is a program for unary arithmetic, based on the classical definitions of addition and multiplication:

$$s \ x + y = s \ (x + y), \quad 0 + y = y, \quad s \ x \times y = (x \times y) + y, \quad 0 \times y = 0.$$

It is essential to understand that the principal port of $+$ is not the output, but the first input. This comes from the fact that $+$ is defined by induction on its first argument. Of course, the same holds for $\times$, and the general interpretation of ports is the following:



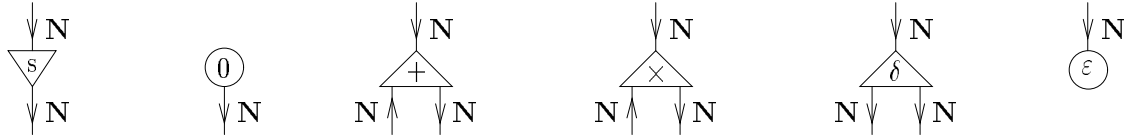
The *duplicator* δ and the *eraser* ε are needed for the multiplication: They are reminiscent of the contraction and the weakening of the previous example.

Our interaction nets provide a simple model of parallel computation. A cell is an agent which is activated only when its principal port is connected to the principal port of another agent. Of course, the two agents must interact simultaneously, but no synchronization is needed between interactions. It is not the case of other models such as cellular automata, where a global synchronization is needed. It is also interesting to notice that our interaction nets are an instance of the *connection graphs*, which were designed as an assembly language for parallel machines (see [Baw86]). Moreover, the

⁶In [GAL92b], another encoding of exponential boxes is proposed, which is inspired by the geometry of interaction and the optimal reduction of lambda-calculus (see [GAL92a]).

sequential models of computation, such as Turing machines, can be revisited in the light of this new paradigm which abolishes the traditional distinction between processor and memory, or between functions and data (see [Laf94]).

Now, we could forget the logical origin of these interaction nets, but it happens that the notions introduced for proof-theoretical purposes have great computational import. Indeed, if the reduction of an interaction net terminates, and no clash or vicious circle appears, then its reduced form is *ready to interact* in the following sense: At least one free port is connected to a principal port of a cell or to another free port. On the other hand, clashes and vicious circles are configurations of deadlock which must be avoided in any computation. A typing discipline is clearly suitable for clashes, but some geometrical criterion is also needed for vicious circles. In the case of our interaction system for unary arithmetic, we can introduce a single type $\mathbf{N}$ and the following typing constraints:



It is easy to check that typed nets are closed under reduction. Since a clash is not typable, reducing a typed net will never create clashes. We can also introduce a switch for δ , and say that a net π is *weakly well-formed* if every switching of π is an acyclic (not necessarily connected) graph. A case analysis shows that this condition is preserved by reduction, and since a vicious circle induces a cyclic switching, reducing a weakly well-formed net will never create vicious circles. In fact, this notion of weakly well-formed net, corresponds to the variant of linear logic considered in section 7. It is equivalent to the notion of *semi-simple net* of [Laf90].

10 Conclusion

The notion of exponential proof-net is not completely convincing because of its hybrid character. The exponential box is indeed a direct translation of the promotion rule of sequent calculus, which is essentially nonlocal. There are two ways to cope with this problem. The first one is the *geometry of interaction*, which consists in replacing nets by more abstract objects (*e.g.* operators in an Hilbert space) in such a way that the cut-elimination can be localized in all cases. The second one has been sketched in the previous section. It consists in replacing the deduction rules of linear logic by interaction rules, in such a way that the concrete character of the computation is preserved, even if its logical status becomes less obvious. In fact, it is not necessary to consider arbitrary systems: In [Laf94], we prove that any interaction system can be translated into a very simple one with only three symbols that we call *interaction combinators*. This means that any computation can be decomposed into elementary steps using a very small number of interaction rules.

There is much more to be done with interaction nets, concerning the theory as well as its applications to computer science. We are thinking of applications to the following areas: programming languages for parallel computation, protocols of communication between interactive softwares, partial evaluation of programs and design of parallel computers.

References

- [Abr93] **S. Abramsky.** *Computational interpretations of Linear Logic*. Theoretical Computer Science **111** 3–57. 1993.
- [Baw86] **A. Bawden.** *Connection Graphs*. Proceedings of ACM Conference on Lisp and Functional Programming, 258–265. 1986.
- [Dan90] **V. Danos.** *La Logique Linéaire appliquée à l'étude de divers processus de normalisation (principalement du λ -calcul)*. Thèse de doctorat, Université de Paris 7. 1990.
- [DR89] **V. Danos & L. Regnier.** *The structure of multiplicatives*. Archive for Mathematical Logic **28**, 181–203. 1989.
- [FR90] **A. Fleury & C. Retoré.** *The Mix rule*. Prépublication de l'équipe de logique 11, Université de Paris 7, to appear in Mathematical Structures in Computer Science. 1990.
- [Gal91] **J. Gallier.** *Constructive Logics. Part II: Linear Logic and Proof Nets*. Digital PRL research report 9. 1991.
- [GAL92a] **G. Gonthier, M. Abadi & J.-J. Levy.** *The geometry of optimal lambda reduction*. Proceedings of 19th ACM Symposium on Principles of Programming Languages (POPL 92). 1992.
- [GAL92b] **G. Gonthier, M. Abadi & J.-J. Levy.** *Linear Logic without boxes*. Proceedings of 7th Annual Symposium on Logic in Computer Science (LICS 92). 1992.
- [Gir87] **J.Y. Girard.** *Linear logic*. Theoretical Computer Science **50**, 1–102. 1987.
- [GLT89] **J.Y. Girard, Y. Lafont & P. Taylor.** *Proofs and Types*. Cambridge Tracts in Theoretical Computer Science **7**, Cambridge University Press. 1989.
- [Laf90] **Y. Lafont.** *Interaction Nets*. Proceedings of 17th ACM Symposium on Principles of Programming Languages (POPL 90) 95–108. 1990.
- [Laf94] **Y. Lafont.** *The Paradigm of Interaction*. In preparation.
- [LW94] **P. Lincoln & T. Winkler.** *Constant-Only Multiplicative Linear Logic is NP-Complete*. To appear in Theoretical Computer Science. 1994.
- [Mac94] **I. Mackie.** *The Geometry of Implementation (an investigation into using the Geometry of Interaction for language implementation)*. Thesis, Imperial College, London. In preparation.
- [Reg92] **L. Regnier.** *Lambda-calcul et réseaux*. Thèse de doctorat, Université de Paris 7. 1992.